# COMMON DENOMINATOR FOR FINITE FAMILIES OF MORPHISMS IN A LOCALIZATION

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*Editorial note.* Counted span is the author's own proof environment, source lines 4496-4514, following the statement at 4484-4494.

*Editorial note.* The proof is complete with no omission markers.

*Allowed standard prerequisites (editorial).*

- Left multiplicative system axioms. definition-multiplicative-system, cited at proof line 4510. Supplies the square-completion axiom (lift with t_2 in S) used to merge two denominators at proof lines 4503-4511.
- Morphisms of the localized category as fractions. the fraction representation of morphisms of S^{-1}C fixed before the statement. Lets each g_i be represented by a pair (X_i -> Y_i, s_i : Y -> Y_i) at proof lines 4497-4498.

*Source reference.* Original definition 04VC: sources/stacks-categories/pinned-original-source.tex, source lines 4123–4123. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Let $\mathcal{C}$ be a category and let $S$ be a left multiplicative system of morphisms of $\mathcal{C}$. Given any finite collection $g_i : X_i \to Y$ of morphisms of $S^{-1}\mathcal{C}$ (indexed by $i$), we can find an element $s : Y \to Y'$ of $S$ and a family of morphisms $f_i : X_i \to Y'$ of $\mathcal{C}$ such that each $g_i$ is the equivalence class of the pair $(f_i : X_i \to Y', s : Y \to Y')$.*

*Proof.* For each $i$ choose a representative $(X_i \to Y_i, s_i : Y \to Y_i)$ of $g_i$. The lemma follows if we can find a morphism $s : Y \to Y'$ in $S$ such that for each $i$ there is a morphism $a_i : Y_i \to Y'$ with $a_i \circ s_i = s$. If we have two indices $i = 1, 2$, then we can do this by completing the square

$$\begin{array}{ccc} Y & \xrightarrow{s_2} & Y_2 \\ {\scriptstyle s_1}\downarrow & & \downarrow{\scriptstyle t_2} \\ Y_1 & \xrightarrow{a_1} & Y' \end{array}$$

with $t_2 \in S$ as is possible by Definition 04VC. Then $s = t_2 \circ s_2 \in S$ works. If we have $n > 2$ morphisms, then we use the above trick to reduce to the case of $n - 1$ morphisms, and we win by induction. □

*Date*: stacks-project:categories.tex:local-snapshot-20261012; GNU FDL 1.2 or later.